



Module 1

Question: Can you explain the importance of ETL in the context of data management?

Answer: ETL, which stands for Extract, Transform, Load, is crucial for several reasons. Firstly, it facilitates the extraction of data from various sources such as databases, applications, and files. Secondly, it enables the transformation of raw data into a format suitable for analysis, which includes cleaning, filtering, and aggregating. Finally, it loads the transformed data into a destination such as a data warehouse or a database for storage and analysis. Without ETL, organizations would struggle to integrate data from disparate sources

Question: What are the key differences between OLAP and OLTP databases?

Answer: OLAP (Online Analytical Processing) and OLTP (Online Transaction Processing) databases serve different purposes in the world of data management. OLTP databases are optimized for handling transactional operations, such as inserting, updating, and deleting records in a database. On the other hand, OLAP databases are designed for complex queries and data analysis, typically involving aggregation, slicing, and dicing of data for reporting and decision-making

purposes. OLAP databases are structured to provide fast query response times for analytical queries, while OLTP databases prioritize fast transaction processing

Question: Describe the process of data warehousing and its role in supporting business intelligence and decision-making.

Answer: Data warehousing involves the collection, storage, and management of data from various sources to facilitate reporting, analysis, and decision-making. In this process, data is extracted from operational systems, transformed into a consistent format, and loaded into a central repository known as a data warehouse. Data warehousing enables organizations to perform complex queries, generate reports, and derive insights from historical and aggregated data, supporting business intelligence initiatives and strategic decision-making processes.